

Savitribai Phule Pune University

T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination

Lab Course CS-321-VSC-P

Foundation of Artificial Intelligence and Machine Learning

Duration: 3 Hours

Maximum Marks:35

Q1) Write a program to demonstrate the Hill Climbing Algorithm for finding the maximum value of a simple objective function. **[10 marks]**

Q2) Write a program to implement the Forward Chaining inference mechanism. **[20 marks]**

OR

Q2) Write a program to implement the Decision Tree Classifier for a classification problem. Use the dataset provided with columns: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age, Outcome(is the class label / target variable: 0 or 1). **[20 marks]**

Pregnancies	Glucose	BP	SkinThick	Insulin	BMI	DiabetesPF	Age	Outcome
6	148	72	35	0	33.6	0.627	50	1
1	85	66	29	0	26.6	0.351	31	0
8	183	64	0	0	23.3	0.672	32	1
1	89	66	23	94	28.1	0.167	21	0
0	137	40	35	168	43.1	2.288	33	1
5	116	74	0	0	25.6	0.201	30	0
3	78	50	32	88	31	0.248	26	1
10	115	0	0	0	35.3	0.134	29	0
2	197	70	45	543	30.5	0.158	53	1
8	125	96	0	0	0	0.232	54	1
4	110	92	0	0	37.6	0.191	30	0

Split the dataset into 70:30 – train:test.

Evaluate the model using `accuracy_score()`. Optimize the model's performance by tuning the criterion (default is GINI, use entropy) and `max_depth` (=2, 3 or 4) parameter of the DT Classifier

Q3) Viva **[05 marks]**